

Bachelor's thesis

HMM for HFT

Predicting and trading on short-term regimes in currency tick data

Egil Furnes
egilsf@uio.no

Department of Mathematics
Faculty of Mathematics and Natural Sciences

11th May 2026

Abstract

Acknowledgements

I would like to thank my supervisor, Associate Professor Gudmund Horn Hermansen (Department of Mathematics, University of Oslo), for his guidance and support throughout this project.

1 Introduction

1.1 The quantitative revolution

Modern financial markets are increasingly mathematical objects. What was once organized around human brokers, trading pits, voice execution, and discretionary judgement has gradually become an ecosystem of electronic limit-order books, automated market makers, statistical arbitrage engines, and latency-sensitive execution algorithms. This transformation is often described through the rise of high-frequency trading, but its deeper significance is the replacement of narrative-based market interpretation with probabilistic modelling, signal extraction, optimization, and real-time statistical decision-making. In this setting, prices are not merely economic summaries of fundamentals; they are also high-dimensional data streams generated by the interaction of heterogeneous agents, algorithms, inventories, order flows, and information shocks.

The popular account of this transition often begins with the drama of speed: faster cables, co-location, microwave links, and the race to react to information milliseconds before competitors (Lewis 2014; Aldridge 2013). However, from a mathematical perspective, speed is only one dimension of the problem. The more fundamental issue is that financial markets produce noisy, non-stationary time series in which any exploitable structure is weak, transient, and rapidly arbitrated away. This makes quantitative trading a natural domain for applied probability, statistics, stochastic processes, and machine learning. The trader is no longer simply asking whether an asset is “cheap” or “expensive”, but whether a small statistical regularity can be detected, validated, traded, and risk-managed after costs.

A useful symbolic figure in this quantitative revolution is Jim Simons, whose background in mathematics and codebreaking helped define the modern idea of finance as a signal-processing problem rather than a purely economic forecasting problem. The lesson is not that markets are deterministic or easily predictable, but almost the opposite: if markets are noisy, adaptive, and adversarial, then any predictive edge must be discovered through large-scale data analysis, rigorous out-of-sample testing, and careful modelling of uncertainty. This philosophy is closely related to the theme of this project. Instead of trying to forecast long-run currency fundamentals, we study whether short-term statistical structure in currency tick data can be captured by models that explicitly allow the market environment to change through time.

1.2 The microstructure of FX and market makers

The foreign exchange market is one of the most natural arenas in which to study this problem. FX is globally traded, highly liquid, electronically intermediated, and active nearly twenty-four hours a day. At the daily or weekly horizon, exchange rates are often modelled as approximately continuous stochastic processes driven by macroeconomic information, interest-rate expectations, risk sentiment, and international capital flows.

At the tick level, however, the market looks very different. Prices move in small discrete increments; bid and ask quotes update asynchronously; liquidity appears and disappears; and the observed price path becomes a sequence of small jumps, reversals, and microstructural frictions rather than a smooth curve.

This microscopic view is the domain of market microstructure. The central question is no longer only what the “true” value of a currency is, but how trading mechanisms, inventory constraints, asymmetric information, and order flow jointly determine observed prices (O’Hara 1995; Hasbrouck 2007). A market maker, for example, typically provides liquidity by quoting both bid and ask prices, earning compensation through the spread while managing inventory and adverse-selection risk. In a quiet regime, the market maker may be able to provide liquidity profitably as prices oscillate locally. In a stressed or information-driven regime, however, the same strategy may become dangerous: order flow can become one-sided, spreads may widen, and price changes may become directional rather than mean-reverting.

This distinction is crucial for high-frequency modelling. At lower frequencies, Brownian motion and diffusion models offer a powerful first approximation to asset-price dynamics, and the connection between finance and physics becomes visible through random walks, diffusion equations, and stochastic calculus (Bachelier 1900; Black and Scholes 1973; Merton 1973). At the tick level, however, the idealized diffusion picture becomes contaminated by the machinery of trading itself. Bid–ask bounce, discreteness, latency, varying liquidity, and order-book dynamics create patterns that are not well captured by a single stationary Gaussian model. The empirical object studied in this thesis is therefore not a clean theoretical Brownian path, but a noisy microstructural signal in which the statistical properties of returns may change rapidly over time.

Currency pairs are especially interesting in this regard because they often have economic and geographical links. Pairs such as AUD/USD and NZD/USD are exposed to related regional and commodity-market forces; EUR/NOK and EUR/SEK connect two Nordic currencies against the euro; and GBP/USD and EUR/USD share the US dollar leg and are deeply embedded in global macro trading. These relationships make currency pairs natural candidates for statistical arbitrage and pairs trading: rather than taking an outright directional view on a single exchange rate, one can model the relative movement between two related exchange rates and test whether temporary deviations tend to revert.



Figure 1: High frequency microstructure

1.3 Non-stationarity and market regimes

The main statistical difficulty is non-stationarity. A standard time-series model usually assumes that the same parameters govern the full sample, or at least that the estimated relationship is stable enough to remain useful out of sample. In financial markets this assumption is often too strong. Volatility clusters, correlations rise and fall, liquidity varies intraday, and the same trading rule may behave very differently depending on whether the market is calm, trending, news-driven, or illiquid. This is not merely an inconvenience; it is one of the defining mathematical features of financial data (Engle 1982; Cont 2001).

For pairs trading, non-stationarity is particularly damaging. A spread that appears mean-reverting in one period may drift in another. A hedge ratio estimated on a long historical sample may be irrelevant over a shorter local window. A pair may be statistically related in the full sample but too slow-moving to trade after transaction costs, or it may exhibit brief episodes of local mean reversion that are invisible to a static model. Thus, the empirical challenge is not only to detect dependence between two currency pairs, but to determine when this dependence is strong enough, stable enough, and cheap enough to trade.

This motivates the language of market regimes. Informally, a regime is a persistent market state with its own statistical properties. One regime may be characterized by low volatility, tight spreads, and local mean reversion; another by high volatility, weak pair-dependence, and directional price movement. The regime itself is not directly observed. We observe only the emitted data: returns, spreads, rolling volatility, correlations, and trading outcomes. The modelling problem is therefore naturally probabilistic: infer the hidden state from the observable time series, and use that inference to adapt the trading rule.

1.4 The solution in Hidden Markov Models

Hidden Markov Models provide a mathematically structured way to represent this idea. The model assumes that the observed process is driven by an unobserved Markov chain whose state determines the distribution of returns. The Markov assumption is deliberately simple: conditional on the current hidden state, the probability of the next state depends only on the present state and not on the full past. Despite this simplicity, the framework is powerful because it allows persistence, sudden switching, and regime-dependent parameters within a single probabilistic model (Rabiner 1989; Hamilton 1989).

In this project, the Hidden Markov Model is used as a regime filter for high-frequency currency-pair trading. The baseline model captures simple spread-based trading behaviour, while the autoregressive extension allows short-term serial dependence to enter the signal. The Markov-switching autoregressive model then allows the parameters of the return process to depend on the inferred market regime. Intuitively, the model asks whether there are hidden “safe” and “dangerous” states in the tick data, and whether avoiding the dangerous state improves the economic performance of a pairs-trading strategy.

This approach is attractive because it is interpretable. Unlike a black-box classifier, the HMM produces transition probabilities, regime-specific volatilities, regime-specific expected returns, and smoothed state probabilities. These quantities can be inspected directly and connected to financial intuition: a high-variance regime may correspond to unstable market conditions, while a low-variance regime may correspond to a more

favourable environment for mean reversion. In this sense, the HMM is not only a predictive model but also a diagnostic tool for studying how microstructural conditions evolve through time.

At the same time, the project is deliberately empirical and sceptical. The purpose is not to assume that a more complex model must outperform a simpler one. In high-frequency finance, additional model complexity can easily overfit noise, especially when trading costs, bid–ask spreads, and walk-forward validation are taken seriously. The key question is therefore comparative: does a Hidden Markov regime filter add value relative to simpler baseline and autoregressive trading rules, or does it mainly reduce exposure without improving risk-adjusted performance?

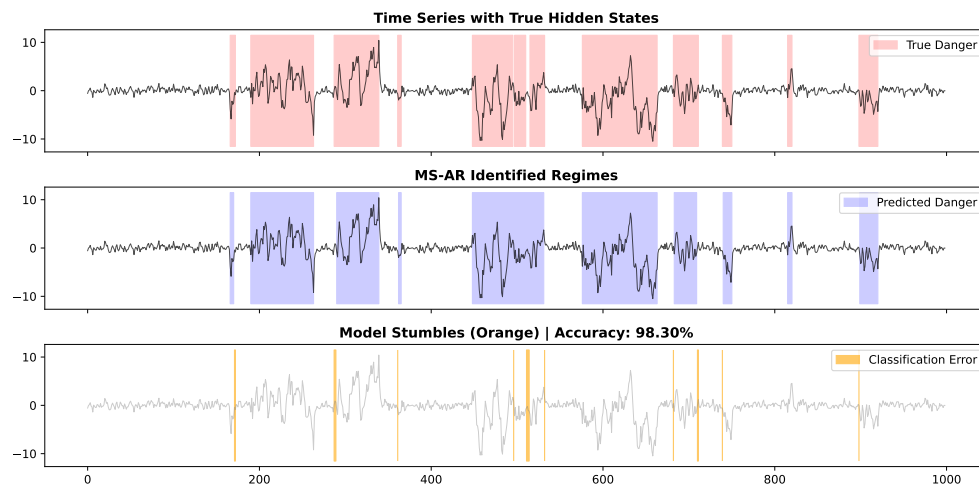


Figure 2: True and hidden states

1.5 Thesis structure

The remainder of the thesis is organized as follows. Section 2 introduces the statistical and financial methodology. It begins with the autoregressive benchmark model, then develops the Hidden Markov and Markov-switching autoregressive framework, before describing the data construction, model-selection procedure, transaction-cost assumptions, and backtesting metrics. Particular emphasis is placed on evaluating performance net of realistic trading frictions, since high-frequency strategies are highly sensitive to spreads and turnover.

Section 3 presents the empirical results for three currency-pair case studies: AUD/USD versus NZD/USD, EUR/NOK versus EUR/SEK, and GBP/USD versus EUR/USD. For each pair, the analysis studies liquidity, return distributions, pair-dependence, cointegration diagnostics, walk-forward optimization, and strategy performance. The baseline, autoregressive, and Markov-switching autoregressive strategies are compared in terms of total return, volatility, drawdown, Sharpe ratio, exposure, win rate, and tail behaviour.

Section 4 concludes by evaluating whether Hidden Markov Models are useful for high-frequency FX pairs trading. The central contribution of the thesis is not a claim that HMMs universally outperform simpler models, but a careful test of when regime awareness helps, when it fails, and how mathematical models behave when confronted with noisy, costly, non-stationary tick data.

2 Methods

In this section we develop the statistical and financial framework used to detect and trade short-term regimes in high-frequency currency-pair data. We begin with the data set and the pre-processing pipeline that turns raw bid–ask tick streams into synchronized observations. We then construct the cointegration spread that drives the trading signal. With the data in place, we introduce the autoregressive baseline, the general Hidden Markov Model, and the Markov-switching autoregressive specification that combines them. We then describe how the model is fit by the EM algorithm, how its hyperparameters are chosen by information criteria, and how the recovered regimes are classified. Finally, we specify the trading strategies, the walk-forward back-testing protocol, and the statistical and financial metrics used to evaluate performance.

2.1 Data

The empirical analysis is based on high-frequency foreign exchange data downloaded from [Dukascopy](#). For each currency pair we collect separate bid and ask quote streams for both legs of the pair. The raw files are stored as monthly parquet files and loaded into [Python through a reproducible data pipeline](#). The sample covers the period from January 2024 to December 2025, giving two full calendar years of tick-level FX observations.

The three currency-pair constructions used in the project are

$$\text{AUD/USD–NZD/USD}, \quad \text{EUR/NOK–EUR/SEK}, \quad \text{GBP/USD–EUR/USD}.$$

These pairs were selected because they represent economically related exchange rates while differing substantially in liquidity, transaction costs, and empirical dependence. AUD/USD and NZD/USD are both Asia-Pacific currencies quoted against the US dollar. EUR/NOK and EUR/SEK connect two Nordic currencies against the euro but are less liquid and have materially wider spreads. GBP/USD and EUR/USD are highly liquid major pairs sharing the US dollar leg.

Using bid and ask data rather than only mid-prices is important in a high-frequency setting. At low frequencies, the spread may be treated as a small implementation detail, but at the tick level it is central to the economics of the strategy. A model may identify statistically significant mean reversion and still fail economically if the expected move is smaller than the cost of trading. For this reason the data set is constructed to preserve bid prices, ask prices, mid-prices, and spread-based transaction-cost estimates throughout the pipeline.

2.2 Pre-processing

The raw Dukascopy quote data are not used directly in the statistical models. Instead, they are transformed into synchronized pair-level observations through a custom Python pipeline. This step is important because high-frequency financial data are irregularly spaced, asynchronous, and strongly affected by bid–ask spreads and microstructure noise. The main pre-processing operations are described below.

2.2.1 Bid–ask and mid prices

For each currency leg, bid and ask quote streams are first sorted by timestamp. Since bid and ask updates do not necessarily arrive at exactly the same time, they are aligned

using a backward as-of merge: at each timestamp, the most recent available quote is used, which avoids any use of future information.

For asset $x \in \{A, B\}$ we then synthesize a mid-price at every tick as

$$M_t^x = \frac{B_t^x + A_t^x}{2}, \quad (1)$$

where B_t^x and A_t^x are the bid and ask quotes of asset x at time t . The mid-price produces a single representative price series for each asset while still preserving the original bid and ask quotes for transaction-cost calculations.

2.2.2 Bid–ask spread

The bid–ask spread is central in HFT because expected profits are often small relative to trading costs. For each asset we measure the half-spread in basis points as

$$\text{HS}_t = \frac{1}{2} \frac{A_t - B_t}{M_t} \times 10^4. \quad (2)$$

The half-spread approximates the cost of crossing the market for one leg of the trade. Since pairs trading requires trading both legs, the combined spread cost is an essential component of the back-test in Section 2.11.

2.3 Cointegration spread

Two non-stationary log-price series $\log A_t$ and $\log B_t$ are said to be cointegrated if there exists a hedge ratio β such that the linear combination

$$S_t = \log A_t - \beta \log B_t \quad (3)$$

is stationary. The hedge ratio β can be estimated globally by the Engle–Granger procedure, which combines an Ordinary Least Squares (OLS) regression with an Augmented Dickey–Fuller (ADF) test on the residuals to verify stationarity. In a high-frequency context, however, β is unlikely to remain constant over the full sample. We therefore model the cointegrating relationship as a slowly varying linear regression and estimate β_t by rolling OLS with window length W_β :

$$\beta_t = \frac{W_\beta \sum_s x_s y_s - (\sum_s x_s)(\sum_s y_s)}{W_\beta \sum_s x_s^2 - (\sum_s x_s)^2}, \quad (4)$$

$$\alpha_t = \frac{1}{W_\beta} \left(\sum_s y_s - \beta_t \sum_s x_s \right), \quad (5)$$

where the sums run over $s \in \{t - W_\beta + 1, \dots, t\}$ and $x_s = \log B_s$, $y_s = \log A_s$.

The cointegration spread at t is then

$$S_t = \log A_t - \beta_t \log B_t - \alpha_t, \quad (6)$$

and the spread return is

$$\Delta S_t = r_t^A - \beta_{t-1} r_t^B, \quad (7)$$

where $r_t^A = \log A_t - \log A_{t-1}$ and similarly for r_t^B . The lag in β_{t-1} ensures the hedge ratio is observable at the time the return is realized.

For the baseline strategy, which makes no use of regime information, the trading signal is the rolling z -score of the spread over a window of length W_z :

$$Z_t = \frac{S_t - \mu_t}{\sigma_t}, \quad \mu_t = \frac{1}{W_z} \sum_{s=t-W_z+1}^t S_s, \quad \sigma_t^2 = \frac{1}{W_z} \sum_{s=t-W_z+1}^t (S_s - \mu_t)^2. \quad (8)$$

2.4 Autoregressive baseline

When modelling a time series, a fundamental objective is to capture the linear dependence between current and past observations. The autoregressive model of order p , denoted $\text{AR}(p)$, formalizes this intuition by projecting the current value of a variable Y_t as a linear combination of its p most recent past values:

$$Y_t = c + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \cdots + \phi_p Y_{t-p} + \epsilon_t. \quad (9)$$

In the context of financial returns, our benchmark is the simplest non-trivial special case: a single-regime $\text{AR}(1)$ on the returns r_t ,

$$r_t = c + \phi r_{t-1} + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \sigma^2). \quad (10)$$

Here c is the constant intercept (from which the unconditional mean is $c/(1 - \phi)$ for $|\phi| < 1$), ϕ measures the strength and direction of the linear serial dependence, and ϵ_t is a Gaussian white-noise innovation. This model captures short-term serial dependence such as momentum or immediate mean reversion under a strict assumption of structural stability.

2.5 Hidden Markov Models

The main limitation of the single-regime $\text{AR}(1)$ is precisely that assumption of structural stability: the parameters c , ϕ , and σ^2 are taken to be constant. Real markets frequently undergo structural breaks and switch between distinct *regimes*, such as transitioning from a low-volatility mean-reverting environment to a high-volatility directional one.

To capture this we introduce an unobserved discrete-time Markov chain $K_t \in \{1, \dots, K\}$ representing the active regime at time t . The dynamics of K_t are governed by a $K \times K$ transition matrix $\mathbf{P} = (p_{ij})$,

$$p_{ij} = \mathbb{P}(K_t = j \mid K_{t-1} = i), \quad (11)$$

with non-negative entries that satisfy the row-sum constraint

$$\sum_{j=1}^K p_{ij} = 1 \quad \forall i. \quad (12)$$

Conditioning the AR process on the hidden state $K_t = k$ allows the dynamics of the returns to adapt to the prevailing regime. The regime-switching $\text{AR}(1)$ for returns is therefore

$$r_t = \mu_k + \phi_k r_{t-1} + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \sigma_k^2). \quad (13)$$

The parameter μ_k is a regime-specific drift, ϕ_k captures how short-term memory of the market changes with the environment, and σ_k^2 models regime-specific volatility, naturally accommodating the volatility clustering frequently observed in financial time series.

To fully specify the Hidden Markov Model we must estimate the parameter set θ , which encapsulates both the state-transition dynamics and the regime-dependent autoregressive parameters:

$$\theta = \left\{ \pi_0, \mathbf{P}, \mu_k, \phi_k, \sigma_k^2 : k = 1, \dots, K \right\}, \quad (14)$$

where $\pi_0 = (\pi_{0,1}, \dots, \pi_{0,K})^\top$ is the initial state distribution with $\pi_{0,k} = \mathbb{P}(K_1 = k)$, $\mathbf{P}$ dictates regime persistence, and the remaining terms are the regime-localized AR parameters.

2.6 Markov-switching AR on the spread

For pairs trading we apply the Markov-switching framework directly to the cointegration spread S_t rather than to the returns of a single asset. Let $K_t \in \{1, 2\}$ be a latent regime indicator. We assume the joint generative model

$$S_t \mid S_{t-1}, K_t = k \sim \mathcal{N}\left(c^{(k)} + \rho^{(k)} S_{t-1}, (\sigma^{(k)})^2\right), \quad (15)$$

$$\mathbb{P}(K_t = j \mid K_{t-1} = i) = p_{ij}, \quad p_{i1} + p_{i2} = 1, \quad \mathbf{P} = \begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix}. \quad (16)$$

This is an AR(1) on the spread with all three parameters switching: the intercept $c^{(k)}$, the autoregressive coefficient $\rho^{(k)}$, and the innovation variance $(\sigma^{(k)})^2$. When $|\rho^{(k)}| < 1$, the unconditional regime mean is

$$\mu^{(k)} = \frac{c^{(k)}}{1 - \rho^{(k)}}, \quad (17)$$

which yields the equivalent mean-deviation parameterization

$$S_t = \mu^{(k)} + \rho^{(k)}(S_{t-1} - \mu^{(k)}) + \varepsilon_t^{(k)}, \quad \varepsilon_t^{(k)} \sim \mathcal{N}\left(0, (\sigma^{(k)})^2\right). \quad (18)$$

Values of $\rho^{(k)}$ close to zero correspond to fast mean reversion within the regime, values close to one correspond to slow mean reversion, and values close to unity correspond to near-random-walk behaviour.

Before estimation we apply two pre-processing steps that improve numerical stability. We winsorize the spread at $\text{median}(S) \pm 4 \cdot \text{std}(S)$, and we rescale it by a factor of 10^4 so that the innovation variance is not numerically negligible.

2.7 Parameter estimation

The HMM parameters θ are estimated by maximum likelihood with the Expectation–Maximization (EM) algorithm (Dempster et al. 1977). The procedure alternates between:

- **E-step.** Compute the smoothed regime posteriors

$$\gamma_t(k) = \mathbb{P}(K_t = k \mid S_1, \dots, S_T; \theta) \quad (19)$$

via the forward–backward algorithm.

- **M-step.** Update the regime-specific parameters $(c^{(k)}, \rho^{(k)}, \sigma^{(k)})$ and the transition matrix $\mathbf{P}$ to maximize the expected complete-data log-likelihood given the posteriors.

The two steps are iterated until the log-likelihood change between successive iterations falls below a small tolerance.

2.8 Model selection

After defining the structural family, the immediate empirical question is the choice of hyperparameters—most notably the number of hidden states K and the autoregressive order p . Adding more states or lags will mechanically improve the in-sample likelihood but introduces the risk of overfitting to statistical noise rather than to genuine market structure. We resolve this trade-off by selecting models with information criteria that penalize complexity. Let m be the total number of estimated parameters, T the number of observations, and $\hat{L}$ the maximized likelihood. The Akaike and Bayesian Information Criteria are

$$\text{AIC} = 2m - 2 \ln \hat{L}, \quad \text{BIC} = m \ln T - 2 \ln \hat{L}. \quad (20)$$

The AIC focuses on minimizing the information lost when the candidate model approximates the true data-generating process and is generally effective for optimizing predictive accuracy. The BIC scales its complexity penalty with $\ln T$, which is more aggressive in large samples and mathematically favours the most parsimonious model that adequately explains the data. During the hyperparameter sweep, models are estimated over a grid of (K, p) values and the configuration that minimizes AIC and BIC is retained as the structural choice for the trading simulations.

2.9 Regime classification

Once K has been fixed by model selection and the parameters estimated, the two regimes are labelled by their innovation variance. We define the mean-reverting (MR) regime as the one with lower volatility and the drifting (DR) regime as the one with higher volatility:

$$k_{\text{MR}} = \arg \min_k \sigma^{(k)}, \quad k_{\text{DR}} = \arg \max_k \sigma^{(k)}. \quad (21)$$

The in-sample regime probabilities used by the trading rules are the corresponding smoothed posteriors

$$\gamma_t^{\text{MR}} = \mathbb{P}(K_t = k_{\text{MR}} \mid S_1, \dots, S_T), \quad \gamma_t^{\text{DR}} = 1 - \gamma_t^{\text{MR}}. \quad (22)$$

A natural soft extension is to scale positions directly by the probability of being in the mean-reverting regime,

$$w(K_t) = \mathbb{P}(K_t \in \text{MR} \mid \mathcal{F}_t), \quad (23)$$

where $\mathcal{F}_t$ denotes the information set available at time t . It is also natural to combine this with volatility scaling. In the strategies developed below, we use either a binary regime gate or a probability-weighted entry threshold rather than continuous position sizing.

2.10 Trading strategies

We compare four strategies that share the same cointegration spread S_t and rolling z -score Z_t as inputs but differ in whether and how they use the regime posteriors γ_t^{MR} and γ_t^{DR} .

2.10.1 Buy & Hold

The do-nothing strategy is always long the spread:

$$P_t^{\text{BH}} = +1 \quad \text{for every bar,} \quad (24)$$

earning ΔS_t at each step and paying *no transaction costs*. Its role is diagnostic: if $\sum_t \Delta S_t > 0$ in a given month, the spread drifted in our favour and any active strategy must outperform that passive drift to add value; if $\sum_t \Delta S_t < 0$, the spread drifted against us.

2.10.2 Baseline (*z*-score)

The textbook pairs-trading rule uses two parameters: a quiet entry threshold z_q and an exit threshold z_x . The position is updated bar by bar according to

$$P_t = \begin{cases} +1 & \text{if } P_{t-1} = 0 \text{ and } Z_t < -z_q, \\ -1 & \text{if } P_{t-1} = 0 \text{ and } Z_t > +z_q, \\ 0 & \text{if } P_{t-1} \neq 0 \text{ and } |Z_t| \leq z_x, \\ P_{t-1} & \text{otherwise.} \end{cases} \quad (25)$$

A long position is opened when the spread is sufficiently below its rolling mean and closed when it returns toward zero; the short rule is symmetric.

In words, for a pair such as AUD/USD–NZD/USD the rule reads

$$\begin{cases} Z_t > +z_q \Rightarrow \text{short AUD/USD and long NZD/USD,} \\ Z_t < -z_q \Rightarrow \text{long AUD/USD and short NZD/USD,} \\ |Z_t| \leq z_x \Rightarrow \text{close the position.} \end{cases} \quad (26)$$

For every unit of the first leg we trade $\hat{\beta}$ units of the second leg, so that the resulting portfolio is exposed to the relative movement between the two currencies rather than to the overall level of the FX market.

2.10.3 Autoregressive (AR) with regime gate

The AR strategy uses the same entry/exit *directions* as the Baseline but adds a binary regime gate

$$G_t = \mathbb{1} \left\{ \gamma_t^{\text{MR}} \geq 1 - \delta \right\}, \quad (27)$$

where $\delta \in (0, 1)$ is a danger-threshold parameter. The modified position rule is

$$P_t = \begin{cases} 0 & \text{if } G_t = 0 \text{ (panic liquidation, no new entries),} \\ \text{baseline rule} & \text{if } G_t = 1. \end{cases} \quad (28)$$

The kill switch is binary: we either trade normally or sit in cash, with no middle ground. Moreover, the kill switch applies to existing positions, not only to new entries—if we are long the spread and the HMM suddenly flags danger, we close out and go flat at the next bar.

2.10.4 Markov-Switching AR (MS-AR)

The MS-AR strategy uses the same entry/exit directions as the Baseline but lets the entry threshold itself be dynamic and probability-weighted. With z_v a wider “volatile entry threshold” that is active in the drifting regime, the effective entry threshold is

$$z_q^{\text{eff}}(t) = \gamma_t^{\text{MR}} z_q + \gamma_t^{\text{DR}} z_v. \quad (29)$$

The position rule is then

$$P_t = \begin{cases} +1 & \text{if } P_{t-1} = 0 \text{ and } Z_t < -z_q^{\text{eff}}(t), \\ -1 & \text{if } P_{t-1} = 0 \text{ and } Z_t > +z_q^{\text{eff}}(t), \\ 0 & \text{if } P_{t-1} \neq 0 \text{ and } |Z_t| \leq z_x, \\ P_{t-1} & \text{otherwise.} \end{cases} \quad (30)$$

When the MR regime is highly probable, the threshold collapses to the tighter z_q and the strategy trades like the Baseline; when the DR regime dominates, the threshold widens toward z_v , requiring a larger deviation before entering.



Figure 3: Long-short positioning in pairs trading.

2.10.5 Strategy summary

Table 1: Strategy definitions and HMM usage.

Strategy	Always long?	Entry threshold (E_t)	Gate (G_t)	Reads HMM?
Buy & Hold	Yes	—	—	No
Baseline	No	z_q	1	No
AR	No	z_q	$\mathbb{1}\{\gamma_t^{\text{MR}} \geq 1 - \delta\}$	Yes, binary
MS-AR	No	$\gamma_t^{\text{MR}} z_q + \gamma_t^{\text{DR}} z_v$	1	Yes, continuous

2.11 Back-testing

Back-testing is used to evaluate whether the model signals translate into economically meaningful trading performance. This step is important because a model can fit

historical returns well without producing a tradable strategy. In high-frequency trading, expected profit per trade is often small, so even modest execution costs can eliminate an apparent statistical edge.

For this reason, the back-test includes realistic market frictions. The most important is the bid–ask spread itself: a trader generally buys at the ask and sells at the bid, rather than trading for free at the mid-price. Commissions and other execution costs are also included, since frequent position changes let small costs accumulate quickly. Financing and margin costs are relevant whenever the strategy holds short or leveraged positions.

The strategies are evaluated in a walk-forward structure, where parameters are estimated on past data and then applied to later out-of-sample periods. This protocol avoids look-ahead bias and provides a more realistic test of whether the Baseline, AR, and Markov-switching strategies would have worked under implementable trading conditions.

2.12 Performance metrics

To rigorously evaluate the trading strategies generated in the simulation we rely on a robust set of quantitative metrics. Simply observing the final account balance is insufficient; a proper evaluation requires us to understand the path taken to achieve those returns, the risks assumed along the way, and the underlying distributional behaviour of the system. These indicators allow us to objectively compare strategies, identifying their specific structural strengths, vulnerabilities, and overall viability in a live market context. Throughout this subsection we use the following notation:

- R_t : return of the asset or strategy at time t ;
- $\mu = \mathbb{E}[R]$: mean return;
- σ : standard deviation of returns;
- N : total number of periods (or trades);
- V_t : portfolio value at time t , with running peak $P_t = \max_{s \leq t} V_s$;
- R_f : a benchmark “risk-free” rate.

2.12.1 Statistical metrics

Statistical metrics describe the shape, spread, and tail behaviour of the return distribution, exposing risks that an equity curve alone can hide. The *annual volatility* σ_{ann} gives a baseline measure of yearly fluctuation, while *skewness* S and *kurtosis* K describe distributional asymmetry and tail fatness:

$$\sigma_{\text{ann}} = \sigma\sqrt{T}, \quad S = \frac{\mathbb{E}[(R - \mu)^3]}{\sigma^3}, \quad K = \frac{\mathbb{E}[(R - \mu)^4]}{\sigma^4}. \quad (31)$$

Tail risk is then summarised by the *Value at Risk* VaR_α at confidence level α , its conditional counterpart CVaR_α , and the *tail factor* TF that compares them:

$$\text{VaR}_\alpha = -Q_{1-\alpha}(R), \quad \text{CVaR}_\alpha = -\mathbb{E}[R \mid R \leq -\text{VaR}_\alpha], \quad \text{TF} = \frac{\text{CVaR}_\alpha}{\text{VaR}_\alpha}. \quad (32)$$

2.12.2 Financial metrics

Financial metrics translate the distributional properties of returns into capital-based outcomes by evaluating the equity curve directly—how efficiently capital is deployed, how the strategy navigates historical losses, and how profitable it is in absolute terms. *Total return* TR is the overall cumulative growth; *annual return* AR standardizes it to a yearly figure (with Y the number of years); *positive months* N_{pos} counts profitable periods:

$$\text{TR} = \prod_{t=1}^N (1 + R_t) - 1, \quad \text{AR} = (1 + \text{TR})^{1/Y} - 1, \quad N_{\text{pos}} = \sum_{t=1}^N \mathbb{1}\{R_t > 0\}. \quad (33)$$

The *drawdown family* captures the worst peak-to-trough drop MDD, the longest recovery time MDD_{dur}, and the Ulcer index UI that combines depth and duration (with $D_t = (P_t - V_t)/P_t$):

$$\text{MDD} = \max_t \left(\frac{P_t - V_t}{P_t} \right), \quad \text{MDD}_{\text{dur}} = \max(t_{\text{recovery}} - t_{\text{peak}}), \quad \text{UI} = \sqrt{\frac{1}{N} \sum_{t=1}^N D_t^2}. \quad (34)$$

Risk-adjusted ratios relate average excess return to a measure of risk: the Sharpe ratio penalizes all volatility, the Sortino ratio penalizes only downside volatility σ_d , and the Calmar ratio uses the maximum drawdown,

$$\text{SR} = \frac{\mu - R_f}{\sigma}, \quad \text{Sortino} = \frac{\mu - R_f}{\sigma_d}, \quad \text{Calmar} = \frac{\text{AR}}{\text{MDD}}. \quad (35)$$

The *profit factor* PF compares total gains to total losses, while the *payoff factor* compares the average sizes:

$$\text{PF} = \frac{\sum \text{Gross Profits}}{\sum |\text{Gross Losses}|}, \quad \text{Payoff} = \frac{\text{Average Win}}{|\text{Average Loss}|}. \quad (36)$$

Finally, trade-level diagnostics summarize the frequency and size of trades. The *win rate* W measures signal accuracy, *market exposure* measures the fraction of time capital is deployed, and *average win* AW and *average loss* AL describe the typical magnitudes:

$$W = \frac{N_{\text{wins}}}{N_{\text{trades}}}, \quad \text{Exposure} = \frac{T_{\text{invested}}}{T_{\text{total}}}, \quad \text{AW} = \frac{1}{N_{\text{wins}}} \sum_{i=1}^{N_{\text{wins}}} \text{Profit}_i, \quad \text{AL} = \frac{1}{N_{\text{losses}}} \sum_{i=1}^{N_{\text{losses}}} \text{Loss}_i. \quad (37)$$

3 Results

4 Conclusion

References

- Aldridge, Irene (2013). *High-Frequency Trading: A Practical Guide to Algorithmic Strategies and Trading Systems*. 2nd ed. Hoboken, NJ: Wiley.
- Bachelier, Louis (1900). ‘Théorie de la spéculation’. In: *Annales scientifiques de l’École Normale Supérieure* 17, pp. 21–86.
- Black, Fischer and Myron Scholes (1973). ‘The Pricing of Options and Corporate Liabilities’. In: *Journal of Political Economy* 81.3, pp. 637–654.
- Cont, Rama (2001). ‘Empirical Properties of Asset Returns: Stylized Facts and Statistical Issues’. In: *Quantitative Finance* 1.2, pp. 223–236.
- Dempster, Arthur P., Nan M. Laird and Donald B. Rubin (1977). ‘Maximum Likelihood from Incomplete Data via the EM Algorithm’. In: *Journal of the Royal Statistical Society: Series B* 39.1, pp. 1–38. DOI: [10.1111/j.2517-6161.1977.tb01600.x](https://doi.org/10.1111/j.2517-6161.1977.tb01600.x). URL: <https://doi.org/10.1111/j.2517-6161.1977.tb01600.x>.
- Engle, Robert F. (1982). ‘Autoregressive Conditional Heteroscedasticity with Estimates of the Variance of United Kingdom Inflation’. In: *Econometrica* 50.4, pp. 987–1007.
- Hamilton, James D. (1989). ‘A New Approach to the Economic Analysis of Nonstationary Time Series and the Business Cycle’. In: *Econometrica* 57.2, pp. 357–384.
- Hasbrouck, Joel (2007). *Empirical Market Microstructure: The Institutions, Economics, and Econometrics of Securities Trading*. Oxford: Oxford University Press.
- Lewis, Michael (2014). *Flash Boys: A Wall Street Revolt*. New York: W. W. Norton & Company.
- Merton, Robert C. (1973). ‘Theory of Rational Option Pricing’. In: *The Bell Journal of Economics and Management Science* 4.1, pp. 141–183.
- O’Hara, Maureen (1995). *Market Microstructure Theory*. Cambridge, MA: Blackwell.
- Rabiner, Lawrence R. (1989). ‘A Tutorial on Hidden Markov Models and Selected Applications in Speech Recognition’. In: *Proceedings of the IEEE* 77.2, pp. 257–286.

5 Appendix